

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Paper No. CHB-602: Inorganic Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 which is compulsory.

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) Why are lower oxidation states more stable in 3d metals and higher oxidation states are more stable in 4d and 5d metals?
- (b) Which of the phenomena among paramagnetism, diamagnetism, and ferromagnetism does not occur in liquid or gaseous state and why?
- (c) Explain the spin-orbit coupling conditions for 3d-metal ions.
- (d) Differentiate between magnetically diluted and magnetically non-diluted substances.
- (e) Using an Orgel diagram, assign electronic transitions to the absorption peaks for:
 - (i) $[\text{CoCl}_4]^{2-}$: 3300 cm^{-1} , 5800 cm^{-1} , 15000 cm^{-1}
 - (ii) $[\text{V}(\text{H}_2\text{O})_6]^{3+}$: 17400 cm^{-1} , 25000 cm^{-1} , 37000 cm^{-1}

Question 2.

- (a) Write the electronic configurations for the 5f series of elements (actinides) and compare their oxidation states with those of the corresponding lanthanides. [6]
- (b) Discuss the separation of Lanthanides by the solvent extraction method. [8]

Question 3.

- (a) Explain the magnetic properties of tetrabutylammonium copper(II) chromium(III) oxalate. How does magnetic coupling occur? [6]
- (b) Calculate the spin-only and experimentally observed magnetic moments of the following ions: Fe^{3+} , Co^{2+} , Ni^{2+} , and Cu^{2+} . [8]

Question 4.

- (a) Compare complex formation tendencies of actinides with those of lanthanides. Give examples. [6]
- (b) Discuss the biological role of nitrogenase enzyme in nitrogen fixation. [8]

Question 5.

- (a) Describe the phenomenon of superexchange where individual magnetic dipoles of metal ions couple through the electrons of intervening ligands. [8]
- (b) How can Ce and Eu be separated from a mixture of other lanthanides using selective oxidation/reduction methods? [6]